

Subject Description Form

Subject Code	AAE3002
Subject Title	Aircraft Structures and Materials
Credit Value	3
Level	3
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: ENG2001 Fundamentals of Materials Science and Engineering AND ME23001 Engineering Mechanics
Objectives	1. To provide students with the key knowledge relevant to aircraft structures and materials; and 2. To provide students with an overview of the composite materials used in modern aircraft; and 3. To provide students with the numerical methods for formulating and solving engineering problems related to aircraft structures and materials.
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Demonstrate a good understanding of the key aspects of aircraft structures; and b. Comprehend the characteristics of the various materials used in aircrafts; and c. Analyze and assess aircraft structures subject to various types of loading using stress analysis methods and failure criteria; and d. Understand the mechanical behaviors of the composite materials used in aircrafts.
Subject Synopsis/ Indicative Syllabus	Characteristics of Aircraft Structures – Aircraft structural elements. Wing, fuselage, tail and landing gear. Riveting, fastener and adhesive joint. Aircraft Structural Analysis – Typical loadings applied on aircrafts. Bending, shear, torsion of thin-wall structures. Structural idealization. Practical stress analysis of wings and fuselages under combined loading. Fundamentals of Aircraft Materials – Material fundamentals. Metallic alloys and their heat treatment. Composites. Failure Criteria for Isotropic Materials – Strength criteria for brittle materials. Yield criteria for ductile materials. Stress concentration. Fatigue. Fracture. Stability of beams under transverse and axial loads. Fundamentals of Aircraft Composites – Mechanical behaviors of composite materials. Processing and fabrication techniques for aircraft composites.

Teaching/Learning Methodology	Lectures are used to deliver the fundamental knowledge in relation to aircraft structures and materials (outcomes a to d).						
	Teaching / Learning Methodology		Intended subject learning outcomes to be covered				
			a	b	c	d	
	1. Lectures		✓	✓	✓	✓	
	2. Tutorials		✓	✓	✓	✓	
Assessment Methods in Alignment with Intended Learning Outcomes							
	Specific assessment methods/tasks		% weighting		Intended subject learning outcomes to be covered		
					a	b	c
	1. Final examination		60%	✓	✓	✓	✓
	2. Take-home assignments and/or in-class quizzes		40%	✓	✓	✓	✓
	Total		100%				
	Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:						
	Overall Assessment:						
	0.6 × End of Subject Examination + 0.4 × Continuous Assessment						
	Examination is adopted to assess students on the overall understanding and the ability of applying the concepts. It is supplemented by the tests and assignments which provide timely feedbacks to both lecturers and students on various topics of the syllabus.						
Student Study Effort Expected	Class contact:						
	▪ Lecture			26 Hrs.			
	▪ Tutorials			13 Hrs.			
	Other student study effort:						
	▪ Self-study			39 Hrs.			
	▪ Assessment preparation			39 Hrs.			
	Total student study effort			117 Hrs.			

Reading List and References	1. C.T. Sun, Mechanics of Aircraft Structures, John Wiley & Sons, latest edition. 2. T.H.G. Megson, Aircraft Structures for Engineering Students, Elsevier, latest edition. 3. R.F. Gibson, Principles of Composite Material Mechanics, McGraw-Hill International Editions, latest edition.
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